MILITARY SPECIFICATION

DETONATOR, STAB, PA501
PARTS FOR,
LOADING, ASSEMBLING AND PACKING

This specification is approved for use by Picatinny Arsenal, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the parts for, loading, assembling and packing for one type of stab detonator designated as Detonator, Stab PA501.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation-Packaging, Methods of
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-650 - Explosives: Sampling Inspection and Testing
MIL-STD-1169 - Packaging, Packing and Marking For Shipment of Inert Ammunition Components

FCS: 1375

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DRAWINGS

PICATINNY ARSENAL

PRODUCT AND PACKING DRAWINGS

9298113 - Detonator, Stab PA501
9258182 - Box, Set-Up, Packing, Ammunition, Inner For Small Explosive Components
9258183 - Box, Set-Up, Packing, Ammunition, Outer For Small Explosive Components
9258184 - Box, Packing, Ammunition, For Small Explosive Components

INSPECTION EQUIPMENT DRAWINGS

8816367 - Firing Pin Guide
8816368 - Disc Lead
8827863 - Sleeve
8827934 - Ball, Steel
9200012 - Firing Pin
9200390 - Firing Assembly
9200602 - Bushing

3. REQUIREMENTS

3.1 Materials.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Detonator.-The detonator shall comply with all requirements specified on Drawing (dwg.) 9298113, and with all requirements specified in applicable specifications.

3.3 Moisture Content

3.3.1 Lead Azide.-The moisture content of the lead azide used in loading the detonator shall not exceed 0.50 percent when tested as specified in 4.5.2.

3.3.2 Primer Mixture.-The moisture content of the primer mixture used in loading the detonator shall not exceed 0.30 percent when tested as specified in 4.5.3.

3.3.3 RDX.-The moisture content of the RDX used in loading the detonator shall not exceed 0.50 percent when tested as specified in 4.5.4.

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3.4 Functioning

3.4.1 Sensitivity and Output.-The detonator shall function on the first attempt and shall cause perforation of the lead disc to a diameter of not less than 0.156 inch when tested as specified in 4.5.7.1.

3.4.2 Waterproofness.-The detonator shall comply with the requirements of 3.4.1 after being submerged in water to a minimum depth of two (2) inches for not less than 48 hours, at a temperature of 75 degrees F. plus or minus 5 degrees, when tested as specified in 4.5.7.2.

3.5 First Article Inspection-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples shall be as specified in the contract.

3.6 Workmanship.-All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility For Inspection and Standard Quality Assurance Provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections.-The following types of inspection shall be conducted on this item.

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

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Part Description	Drawings	Quantity
Disc, Detonator Closing	9298116	25
Disc, Detonator	9298115	25
Cup, Detonator	9298114	25
Detonator, Stab PA501	9298113	340

4.3.2 Inspections to be Performed.-See MIL-A-48078, and Table I specified herein.

4.3.3 Rejection.-(See MIL-A-48078).

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-D-63057 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	PA501 Stab Detonator and Components				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Disc, Detonator Closing (Dwg. 9298116)				
	Examination for Defects	25	-	3.2	4.4.2.1
	Disc, Detonator (Dwg. 9298115)				
	Examination for Defects	25	-	3.2	4.4.2.2
	Cup, Detonator (Dwg. 9298115)				
	Examination for Defects	25	-	3.2	4.4.2.3
	Detonator, Stab PA501 (Dwg. 9298113)				
	Examination for Defects	(a)	-	3.2	4.4.2.7
	Functioning-Sensitivity and Output	(b)	-	3.4.1	4.5.7.1
	Functioning-Waterproofness	25	-	3.4.2	4.5.7.2

NOTES: (a) The sample shall consist of the detonators to be functioned per 4.5.7.1 and 4.5.7.2
(b) See 4.5.7.1

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4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of detonators shall contain:

a. Parts bearing the same lot interfix number from one manufacturer.

b. Lead azide of the same type from not more than two lots produced by the same manufacturer, in accordance with the same specification and same specification revision and which have subsequently been subjected to uniformly controlled drying and operating conditions at the loading plant.

c. RDX from not more than one lot.

d. Primer mixture from one batch or blending of batches which results in homogeneity of all primer mixture used in loading the entire detonator lot.

4.4.2 Examination.-(See MIL-A-48078).

a. Sampling Plans.-Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-D-63057 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Disc, Detonator Closing				9298116
					NEXT HIGHER ASSEMBLY 9298113
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter		0.40%	3.2	Gage
102	Thickness		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-D-63057 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Disc, Detonator				9298115
					NEXT HIGHER ASSEMBLY 9298113
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter		0.40%	3.2	Gage
102	Thickness		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63057 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Cup, Detonator				9298114
					NEXT HIGHER ASSEMBLY 9298113
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Cup, split or cracked		100.0%	3.2	Visual
Major					
101	Length		0.40%	3.2	Gage
102	Inside diameter		0.40%	3.2	Gage
103	Outside diameter		0.40%	3.2	Gage
104	Thickness through bottom		0.40%	3.2	Gage
105	Diameter of hole, minimum		0.40%	3.2	Gage
106	Concentricity of hole with outside diameter		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-D-63057 (PA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Detonator, Stab PA501 Prior to loading upper charge	1 OF 1	9298113
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Disc not secure in bottom of cup		0.40%	3.2	Visual/Manual
102	Moisture content of lead azide	(a)(b)		3.3.1	4.5.2
103	Moisture content of primer mixture	(a)(c)		3.3.2	4.5.3
104	Moisture content of RDX	(a)(d)		3.3.3	4.5.4
105	Binder-Lubricant content of RDX	(a)(d)		3.2	4.5.5
106	Primer mix composition	(a)(e)		3.2	4.5.6
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual
NOTES:(a)	If the sample fails the test and loading has not begun, the lot of tested material shall be rejected until proper corrective action has been accomplished as verified by reperforming the test. If loading has begun, action on the unloaded material shall be as specified above and all detonators loaded with the questionable material shall be rejected.				
(b)	A sample of 2 grams shall be selected for each lot of lead azide at the time of loading.				
(c)	A sample of 0.7 gram shall be selected for each lot of primer mixture at the time of loading.				

NOTES: (d) A sample of 5 grams of the RDX and binder-lubricant mixture shall be selected for each lot at the time of loading.
(e) A sample of 0.5 gram shall be selected for each lot of primer mixture at time of loading.

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CLASSIFICATION OF DEFECTS & TESTS MIL-D-63057 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9298113
4.4.2.5	Detonator, Stab PA501 (Prior to loading intermediate charge)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Upper charge weight below minimum		0.40%	3.2	Balance
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63057 (PA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Detonator, Stab PA501 (Prior to loading lower charge)	1 OF 1	9298113
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Intermediate charge weight below minimum		0.40%	3.2	Balance
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-63057 (PA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Detonator, Stab PA501	1 OF 1	9298113
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Cup cracked or split		100.0%	3.2	Visual
Major					
101	Length		0.40%	3.2	Gage
102	Diameter		0.40%	3.2	Gage
103	Disc loose, punctured, torn or wrinkled to such a degree that the contents are exposed		0.40%	3.2	Visual
104	Color of lacquer incorrect on either end		0.40%	3.2	Visual
105	Cup damaged except as described in criteria above		0.40%	3.2	Visual
106	Crimp not full 360 degrees		0.40%	3.2	Visual
107	Functioning-sensitivity and output	(a)		3.4.1	4.5.7.1
108	Functioning-waterproofness	25		3.4.2	4.5.7.2
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

(a) See 4.5.7.1

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.8	Box, set-up, inner, after loading and prior to closing slide cover		9258182
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Filler and cushion assembly missing		100.0%	3.2	Visual
2.	Filler and cushion assembly improperly assembled		100.0%	3.2	Visual
3.	Stiffener missing		100.0%	3.2	Visual
4.	Slide cover assembled reversed		100.0%	3.2	Visual
Major					
101	Release paper missing		0.40%	3.2	Visual
Minor					
201	Marking missing incomplete or illegible		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-D-63057 (PA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Box, set-up, outer, with inner set-up boxes assembled and prior to assembling top and sealing strip	1 OF 1	9258183
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Red surfaces on inner set-up boxes do not alternate on same side of outer set-up box				
2.	Stiffener not facing one direction		100.0%	3.2	Visual
Major					
101	Heat seal test of bag	(a)		3.2	4.5.8
Minor					
201	Marking missing, incomplete or illegible		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

(a) See 4.5.8

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.10	Box, packing, ammunition prior to assembling cover and strapping		9258184
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Bags containing set-up boxes damaged		0.40%	3.2	Visual
102	Cushioning material incomplete or missing from around outer set-up boxes		0.40%	3.2	Visual
103	Marking missing, misleading or illegible		0.40%	3.2	Visual
104	Barrier bag seal incomplete or improperly positioned		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS MIL-D-63057 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.11	Box, packing, ammunition				9258184
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Strapping missing, broken, loose or improperly fastened		0.40%	3.2	Visual
102	Box damaged		0.40%	3.2	Visual
103	Marking missing, incomplete or illegible		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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4.4.3 Testing.-Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed in this specification is identified, either directly or by reference, in the "Paragraph Reference/Inspection Method" column of the First Article or Quality Conformance Inspection Tables herein. The contractor shall submit inspection equipment designs to the Government for approval in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.2 herein.

4.5 Test Methods and Procedures

4.5.1 Materials, Components and Processes.-Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and Mechanical Inspections.-The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.4.2 herein and Table I.

4.5.2 Moisture Content of Lead Azide.-The moisture content of the lead azide shall be determined at the time of loading. The moisture content of the lead azide shall be determined in accordance with MIL-STD-650, Method 101.5 except, use a two gram sample and cover the dish with a sheet of polytetrafluoroethylene.

4.5.3 Moisture Content of Primer Mixture.-The moisture content of the primer mixture shall be determined at the time of loading. A sample of 0.7 gram shall be selected. The sample shall be accurately weighed to the nearest milligram and dried to a constant weight at 60 degrees C. plus or minus 5 degrees C., or in a desiccator containing anhydrous calcium chloride. The loss in weight shall be calculated as the percentage of moisture as follows:

$$\text{Percentage of Moisture} = \frac{100(A-B)}{A}$$

where:

A = The weight of the sample before drying.

B = The weight of the sample after drying.

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4.5.4 Moisture Content of RDX.-The moisture content of the RDX shall be determined at the time of loading. The moisture content of the RDX shall be determined in accordance with MIL-STD-650, Method 101.5.

4.5.5 Binder-Lubricant Content of RDX.-The calcium resinate, graphite, and total binder-lubricant content of the RDX to be used in the detonator shall be determined as follows:

4.5.5.1 Calcium Resinate.-An accurately weighed portion of five (5) grams of the sample shall be transferred to a tared medium porosity sintered-glass crucible. The sample shall be extracted with three 20 milliliter (ml) portions of chloroform which have been previously saturated with RDX, maintaining each portion in contact with the residue for approximately five minutes before applying suction. Any lumps which may persist shall be broken up by stirring occasionally and the sides of the crucible washed down with each 20 ml portion of chloroform to insure complete removal of the calcium resinate. The residue shall be aspirated until the odor of chloroform is no longer detectable; the crucible and contents shall be placed in a drying oven at 95-105 degrees C. for 1/2 hour, cooled in a desiccator and weighed. The above extraction shall be repeated, drying, and weighing with individual 20 ml portions of chloroform until the loss in weight on subsequent extractions totals less than 0.003 grams. The loss in weight of the crucible plus contents shall be calculated to percent calcium resinate in the sample on a moisture free basis as follows:

$$\text{Percentage of Calcium Resinate} = \frac{100A}{W}$$

where:

A = loss in weight of the crucible plus contents.
W = weight of sample on moisture free basis.

4.5.5.2 Graphite.-The crucible and contents remaining after the removal of calcium resinate shall be washed with a minimum of three 20 ml portions of acetone, each portion maintained in contact with the residue for approximately one (1) minute with continuous stirring before applying suction. Repeat washing operation until the following qualitative test for RDX in the acetone washings is negative. The acetone filtrate shall be transferred to a 25 ml beaker, evaporated to dryness on a steam bath and several drops of a solution of chromotropic acid added in 36N sulfuric acid. The formation of a reddish-brown color, as compared to a pink color obtained by a blank prepared in an identical manner indicates the presence of RDX. The residue shall be aspirated until

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the odor of acetone is no longer detectable; the crucible and contents shall be placed in a drying oven at 95-105 degrees C for 1/2 hour, cooled in a desiccator and weighed. The weight of the residue in the crucible shall be calculated to the percentage of graphite in the sample on a moisture free basis as follows:

$$\text{Percentage of Graphite} = \frac{100A}{W}$$

where:

A = increase in weight of filtering crucible.
W = weight of sample on a moisture free basis.

4.5.5.3 Total Binder-Lubricant.-The percentage of total binder-lubricant in the sample shall be calculated by adding the percentage of calcium resinate and graphite found.

4.5.6 Composition of Primer Mixture.-The composition of the primer mixture shall be determined by performing the following tests in the sequence specified:

4.5.6.1 Barium Nitrate.-An accurately weighed portion of approximately 0.5 gram of primer mixture shall be transferred to a tared medium-porosity, fritted glass crucible of 20 ml capacity. Three ml of lead azide saturated distilled water shall be added at a temperature of 50 degrees C plus or minus 2 degrees C, agitated for one (1) minute, and, if necessary, lumps shall be broken up with a rubber policeman attached to a glass rod. Suction shall be applied. This procedure shall be repeated five times, making a total of six, 3-ml extractions. The sample in the filtering crucible shall be rinsed three times with alcohol, dried in an oven maintained at 60 degrees C plus or minus 5 degrees C for 30 minutes, cooled in a desiccator and weighed. The loss in weight shall be calculated as the percentage of barium nitrate in the sample on a dry basis as follows:

$$\text{Barium nitrate, percent} = \frac{(A-B)\ 100}{W(1-.01M)}$$

where:

A = weight of crucible and sample.
B = weight of the crucible and sample after extractions with lead azide saturated water.
M = moisture content of sample obtained from 4.5.3 in percent.
W = weight of sample.

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4.5.6.2 Lead Styphnate

4.5.6.2.1 Basic Lead Styphnate (when applicable).-The residue obtained in the determination of barium nitrate shall be extracted with six 5 ml portions of saturated ammonium acetate solution at about 25 degrees C, (hot saturated ammonium acetate shall not be used as it dissolves antimony sulfide to some extent) agitating each portion for about 30 seconds and allowing each portion to remain in contact with the residue for approximately two minutes. The residue shall be washed with water until the filtrate coming through the crucible appears colorless. Reserve the crucible and residue for the tetracene determination. The filtrate and washings shall be quantitatively transferred to a 250 ml calibrated volumetric flask, the beaker washed thoroughly with distilled water and diluted to the mark. Using a calibrated pipette, a 2 ml aliquot of this solution shall be transferred to a 50 ml calibrated volumetric flask and diluted to the mark with distilled water. A portion of this solution shall be transferred to a corex glass spectrophotometric cell having a width of 1 centimeter (cm) and its absorbance determined at a wave length of 410 millimicrons. As reference, a similar cell containing saturated ammonium acetate solution of the same concentration as the sample solution shall be used. Since the cell which holds the basic lead styphnate is not identical with the reference cell the absorbance obtained as described above shall be corrected for the difference in the amount of light which the two cells scatter and absorb. To do this both cells shall be filled with ammonium acetate solution and the absorbance of the cell which originally contained the basic lead styphnate solution shall be measured at a wave length of 410 millimicrons. The percentage of basic lead styphnate in the sample shall be calculated on a dry basis as follows:

$$\text{Basic lead styphnate, percent} = \frac{F\,(A-B)}{W(1-.01M)D}$$

where:

A = absorbance of basic lead styphnate solution.
B = absorbance of ammonium acetate solution.
D = width of corex cell, cm.
M = moisture content of sample obtained from 4.5.3 in percent.
W = weight of sample.
*F = spectrophotometric factor. To determine the spectrophometric factor (F) weigh a 0.5 gm sample of pure lead styphnate. Transfer the lead styphnate quantitively with 50 to 100 ml of distilled water to a 500 ml volumetric flask,

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add 30 ml of saturated ammonium acetate solution to the solution in the flask and dilute to mark with distilled water. Using aliquots of the solution prepare five solutions of lead styphnate ranging in concentration from .5 to 2.5 mg per 100 ml. Determine the absorbance of each solution at 410 millimicrons. Divide concentrations of lead styphnate for each of the solutions by its corresponding absorbance value. The average of the five quotients thus obtained shall be used as the spectrophometric factor (F) for the lead styphnate.

4.5.6.2.2 Determination of Normal Lead Styphnate (when applicable).-Extract the residue obtained in the barium nitrate determination with sufficient 5 ml portions of ammonium acetate solution (saturated) at about 25°C until the washings are colorless. Agitate each portion for about one minute, then allow each portion to remain in contact with the residue for one additional minute. Finally wash the residue with water, saturated with antimony sulfide, until the washings come through clear. Reserve the cricible and residue for the tetracene determination. Transfer the filtrate and washings quantitatively to a 250 ml calibrated volumetric flask, washing the beaker thoroughly with distilled water and dilute to the mark with distilled water. Using a calibrated pipette, transfer a 2 ml aliquot of this solution to a second 100 ml calibrated volumetric flask and dilute to the mark with distilled water. Transfer a portion of this solution to a corex glass spectrophotometric cell having a width of 1 cm and determine its absorbance at a wave length of 410 millimicrons. Use as reference, a similar cell containing saturated ammonium acetate solution of the same concentration as the sample in the final dilution. Since the cell which holds the normal lead styphnate is not identical with the reference cell, correct the absorbance obtained as described above for the difference in the amount of light which the two cells scatter and absorb. To do this, fill both cells with ammonium acetate solution, and measure at a wave length of 410 millimicrons the absorbance of the cell which originally contained the normal lead styphnate solution. Calculate the percentage of normal lead styphnate in the sample on a dry basis as follows:

$$\text{Normal lead styphnate, percent} = \frac{F\,(A-B)}{WD}$$

where:

A = absorbance of normal lead styphnate solution.
B = absorbance of ammonium acetate solution.
D = width of corex cell, cm.
W = weight of the dried sample.
F = spectrophotometric factor. To determine the spectrophometric factor (F) weigh a 0.5 gm sample of pure lead

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styphnate. Transfer the lead styphnate quantitatively with 50 to 100 ml of distilled water to a 500 ml volumetric flask, add 30 ml of saturated ammonium acetate solution to the solution in the flask and dilute to mark with distilled water. Using aliquots of the solution prepare five solutions of lead styphnate ranging in concentration from .5 to 2.5 mg per 100 ml. Determine the absorbance of each solution at 410 millimicrons. Divide concentrations of lead styphnate for each of the solutions by its corresponding absorbance value. The average of the five quotients thus obtained shall be used as the spectrophometric factor (F) for the lead styphnate.

4.5.6.3 Tetracene.-The wet residue obtained in the determination of lead styphnate shall be washed three times with alcohol, dried in an oven maintained at 60 degrees C plus or minus 5 degrees C for 30 minutes, cooled in a desiccator and weighed. The dried residue in the crucible shall be transferred to a 125 ml beaker with a stream of water, 25 ml of water shall be added, and the contents of the beaker boiled on a hot plate for five minutes. The contents of the beaker shall be poured into the same crucible, and the residue in the crucible washed three times with boiling water, and finally with alcohol. The crucible and contents shall be dried in an oven maintained at 60 degrees C plus or minus 5 degrees C for 30 minutes, cooled in a desiccator and weighed. The loss in weight shall be calculated as the percentage of tetracene in the sample on a dry basis as follows:

$$\text{Tetracene, percent} = \frac{(C-D)\ 100}{W(1-.01M)}$$

where:

C = weight of crucible and residue after extraction with saturated ammonium acetate solution.
D = weight of crucible and residue after treating with hot water.
W = weight of the sample.
M = moisture content of sample obtained from 4.5.3 in percent.

4.5.6.4 Antimony Sulfide.-The weight of the residue shall be calculated from the determination of tetracene to percentage antimony sulfide in the sample on a dry basis as follows:

$$\text{Antimony sulfide, percent} = \frac{(D-E)}{W(1-.01M)}$$

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where:

D = weight of crucible and sample after treating with hot water.
E = weight of crucible.
W = weight of sample.
M = moisture content of sample obtained in 4.5.3 in percent.

4.5.6.5 Lead Azide.-The lead azide content in the sample of a dry basis, shall be determined by subtracting from 100 the combined percentages of barium nitrate, lead styphnate, tetracene and antimony sulfide.

4.5.7 Functioning

4.5.7.1 Sensitivity and Output.-A sample of three hundred and fifteen (315) detonators shall be randomly selected from each lot. If three or more detonators fail to comply with the specified requirements, the lot shall be rejected. If one or two defectives occur, a second sample of 315 detonators shall be selected and tested. If, in the combined first and second sample, a total of four (4) or more defectives occur, the lot shall be rejected. The detonator shall be assembled to the applicable test fixture specified in 4.4.4, so that the firing pin will strike the sensitive end of the detonator. A steel ball 1-3/16 inch in diameter and weighing 3.95 ounces maximum, shall be dropped on the firing pin, to function the detonator, from a height of 3 inches, maximum. The height of drop shall be measured as the height from the top of the firing pin to the bottom of the ball. A detonator that fails to fire or fails to create a hole in the lead disc of the required size shall be classed defective. The inspection equipment shall be in accordance with dwgs 9200390, 8816367, 9200012, 8827934, 8827863, 8816368 and 9200602.

4.5.7.2 Waterproofness.-The detonator shall be submerged in water maintained at a temperature 75 degrees F plus or minus 5 degrees at the required depth for the time specified and within one hour upon removal from water subjected to the sensitivity and output test specified in 4.5.7.1. A detonator that fails to fire or fails to create a hole in the lead disc of the required size shall be classed defective. The lot shall be rejected if one or more defectives occur.

4.5.8 Heat Seal of Sealed Bag.-The adequacy of the seal on the bag shall be determined in accordance with MIL-P-116.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

5.1.1 Level C.-The detonators shall be packaged and marked in accordance with dwgs 9258182 and 9258183.

5.1.2 Level C.-The detonator parts shall be packaged in accordance with MIL-STD-1169 requirements.

5.2 Packing

5.2.1 Level C.-The detonators shall be packed and marked as specified on dwg. 9258184.

5.2.2 Level C.-The detonator parts shall be packed in accordance with MIL-STD-1169 requirements.

5.3 Marking.-Detonator containers and bags shall be marked in accordance with instructions contained in dwgs 9258182, 9258183, and 9258184. Detonator parts containers shall be marked in accordance with MIL-STD-1169 requirements.

6. NOTES

6.1 Ordering Data.-See MIL-A-48078 (PA).

6.2 Submission of Inspection Equipment Designs For Approval.-(See MIL-A-48078). Submit equipment designs as required, to Commander: Picatinny Arsenal, ATTN: SARPA-QA-T Dover, New Jersey 07801.

6.3 Check Test For Possible Deterioration.-The following test is designed to prevent detonators that have deteriorated from being assembled into major ammunition items. The test shall be performed on detonators from each lot which have been stored under normal conditions for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the detonators are assembled does not require that the check test be performed, the Contracting Officer should insert in the contract a provision for the performance of the check test. The Contracting Officer will arrange for the contractor to be reimbursed for expense incurred in his performance of the check test for possible deterioration. Detonator assemblies shall be selected from each lot for this test.

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6.3.1 Output.-A sample of 315 detonators shall be selected at random from each lot. If four (4) or more detonators fail to comply with the requirement specified, the lot shall be rejected. If more than one (1) defective is found, a second sample of 315 shall be selected at random and tested. If in the combined first and second sample a total of five (5) or more defectives are found the lot shall be rejected. The test shall be performed as specified in 4.5.7.1 using test equipment as specified in 4.4.4.

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Preparing Activity:
Army-PA

Project Number: 1375-A-164

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